energy into other forms of energy. In computer based applications, matrices play a vital role in the projection of three dimensional image into a two dimensional screen, creating a realistic seeming motions. In graphic software, Matrix Algebra is used to process linear transformations to render images. One of the most important usage of matrices are encryption of message codes. The encryption and decryption process are carried out using matrix multiplication and inverse operations. The concept of matrices is used in transmission of codes when the messages are lengthy. In Geology, matrices are used for taking seismic surveys. In Robotics, matrices are used to identify the robot movements.

Definition

A matrix is a rectangular array of elements. The horizontal arrangements are called rows and vertical arrangements are called columns.

For example, $\begin{pmatrix} 4 & 8 & 0 \\ 1 & 9 & -2 \end{pmatrix}$ is a matrix.

Usually capital letters such as $A,\ B,\ C,\ X,\ Y,\ \ldots$ etc., are used to represent the matrices and small letters such as $a,\ b,\ c,\ l,\ m,\ n,\ a_{12},\ a_{13},\ \ldots$ to indicate the entries or elements of the matrices.

The following are some examples of matrices

(i) $\begin{pmatrix} 8 & 4 & -1 \\ \frac{1}{2} & 5 & 4 \\ 9 & 0 & 1 \end{pmatrix}$ (ii) $\begin{pmatrix} 1+x & x^3 & \sin x \\ \cos x & 2 & \tan x \end{pmatrix}$ (iii) $\begin{pmatrix} 3+1 & \sqrt{2} & -1 \\ 1.5 & 8 & 9 \\ \frac{1}{3} & 13 & \frac{-7}{9} \end{pmatrix}$

3.9.1 Order of a Matrix

If a matrix A has m number of rows and n number of columns, then the order of the matrix A is (Number of rows) $\times$ (Number of columns) that is, $m \times n$ .We read $m \times n$ as m cross n or m by n. It may be noted that $m \times n$ is not a product of m and n.

General form of a matrix A with m rows and n columns (order $m \times n$) can be written in the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1j} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2j} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mj} & \ldots & a_{mn} \end{pmatrix}$$

where, $a_{11}, a_{12}, \ldots$ denote entries of the matrix. a_{11} is the element in first row, first column, a_{12} is the element in the first row, second column, and so on.

Progress Check

1. Find is the element in the second row and third column of the matrix $\begin{pmatrix} 1 & -2 & 3 \\ 2 & 1 & 5 \end{pmatrix}$
2. Find is the order of the matrix $\begin{pmatrix} \sin\theta \\ \cos\theta \\ \tan\theta \end{pmatrix}$
3. Determine the entries denoted by $a_{11}, a_{22}, a_{33}, a_{44}$ from the matrix $\begin{pmatrix} 2 & 1 & 3 & 4 \\ 5 & 9 & -4 & \sqrt{7} \\ 3 & \frac{5}{2} & 8 & 9 \\ 7 & 0 & 1 & 4 \end{pmatrix}$

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